

Lab 2

Name: Abdulaziz Khaled Alhassan

ID: 2240006848

Group: 7MA1

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In [1]: #import the libraries
import cv2
import numpy as np
import matplotlib.pyplot as plt
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In [2]: #Sampling function
def sample_image(image, factor):

    height, width = image.shape[:2]
    sampled_image = cv2.resize(image, (width // factor, height //
factor), interpolation=cv2.INTER_NEAREST)
    return sampled_image

#Quantization function
def quantize_image(image, levels):

    quantized_image = np.floor(image / (256 // levels)) * (256 //levels)
    quantized_image = quantized_image.astype(np.uint8)
    return quantized_image

#Plotting function
def plot_images(original, sampled, quantized):

    plt.figure(figsize=(12, 4))
    plt.subplot(1, 3, 1)
    plt.imshow(original, cmap='gray')
    plt.title('Original Image')
    plt.axis('off')
    plt.subplot(1, 3, 2)
    plt.imshow(sampled, cmap='gray')
    plt.title('Sampled Image')
    plt.axis('off')
    plt.subplot(1, 3, 3)
    plt.imshow(quantized, cmap='gray')
    plt.title('Quantized Image')
    plt.axis('off')
    plt.show()
```

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In [3]: #Image path and parameters
image_path='lena_gray_256.tif'
sampling_factor=5
quantization_levels=11

#Load image in grayscale
original_image = cv2.imread(image_path, cv2.IMREAD_GRAYSCALE)
if original_image is None:
    print(f"Error: Unable to load image at {image_path}")

#Sample and quantize
sampled_image = sample_image(original_image, sampling_factor)
quantized_image = quantize_image(original_image, quantization_levels)

#Plot results
plot_images(original_image, sampled_image, quantized_image)
```

Original Image



Sampled Image



Quantized Image



```
In [4]: #Import Image from PIL
        from PIL import Image

        #Open the images
        img1=Image.open('lena_gray_256.tif')
        img2=Image.open('cameraman.tif')

        #Change the size of the images to be the same
        resize=(400,400)
        img1=img1.resize(resize,Image.Resampling.LANCZOS)
        img2=img2.resize(resize,Image.Resampling.LANCZOS)

        #Convert the images to numpy arrays
        im1arr=np.asarray(img1)
        im2arr=np.asarray(img2)

        #Task 2.1 : subtract two images and display the result
        subtraction=im1arr-im2arr
        resultImage=Image.fromarray(subtraction)
        plt.show()

        #Task 2.2 : add one image to a constant and display the result
        constant_addition=im1arr+175
        resultImage=Image.fromarray(constant_addition)
        plt.show()

        #Task 2.3 : set difference operation between two gray-scale images and
        #display the result
        set_difference = im1arr - im2arr
        resultImage=Image.fromarray(set_difference)
        resultImage.show()

        #Task 2.4 : symmetric difference operation between two gray-scale images and
        #display the result
        symmetric_difference = np.abs(im1arr - im2arr)
        resultImage=Image.fromarray(symmetric_difference)
        resultImage.show()

        #Adding two images and display the result
        addition=im1arr+im2arr
        resultImage=Image.fromarray(addition)
        resultImage.show()
```

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In [5]: img3 = Image.open('new_image.jpg').convert('L')
img3.show()
img4 = Image.open('new_image2.jpg')
img4.show()
resize = (400, 400)
img3 = img3.resize(resize, Image.Resampling.LANCZOS)
img4 = img4.resize(resize, Image.Resampling.LANCZOS)
im3arr = np.asarray(img3)
im4arr = np.asarray(img4)

#union operation
union = im4arr | im3arr
resultImage2 = Image.fromarray(union)
resultImage2.show()

#task2.5 : intersection of two gray-scale images and display the result
intersection = im4arr & im3arr
resultImage3 = Image.fromarray(intersection)
resultImage3.show()
```